

Turn your health into an adventure.

A pixel-art fantasy RPG that is also a health tracker, built on real science and honest numbers.

Water · Food · Sleep · Stress · Stairs · Running · Heart rate

GROUP 14
MARA · Kolej MARA Kulim

3 UN SDG 3
Good Health and Well-being

Wellness tracker, not a medical device

THE QUEST

Health tracking is easy to start and hard to keep up. In HealthWiz Kingdom every healthy action has a visible effect.

You log **water, meals, sleep, stairs** (with workout heart rate), **runs** and **stress**. **Medius, the Wizard King,** guides you as your real actions restore the kingdom, with **Luna,** an apprentice witch, as your everyday companion.

HOW IT WORKS

1. **Log** a real habit: a glass of water, a meal, a climb.
2. **Restore** the kingdom: its regions heal as you keep going.
3. **Understand:** a "?" shows the formula and sources behind every number.

WHAT SETS IT APART

1. **Health first, game second.** Game values (XP, levels, badges) are tagged "game progress, not a health measurement".
2. **Every number shows its working:** inputs, formula, your numbers, sources, limits.
3. **Honest about uncertainty.** Values are labelled **measured, calculated, estimated, simulated** or **missing**. No invented readings.
4. **Private and optional.** Works on your device with no account; an account is only for backup and sync.

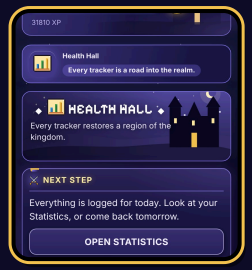
8	8	104	102
REGIONS	THEMES	MODULES	TESTS

SEE IT RUNNING

REAL SCREENS FROM THE LIVE APP



HOME
Today's health at a glance



HEALTH HALL
Every tracker, one next step



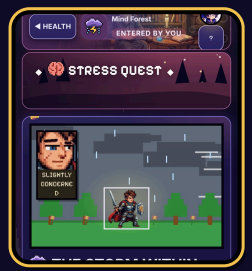
THE KINGDOM
Eight regions, all restored



CAMPUS MAP
3D, with a heart on each staircase



NUTRITION
kcal and mL against your goal



STRESS QUEST
The Storm Within



TRAINING HALL
Pick a goal, muscle and move



SCIENCE LAB
Test a guess with your own logs

Captured from the running app on a phone-sized screen with 30 days of sample entries. The numbers are demo data, not real health measurements.

BEGIN YOUR QUEST

healthwizkingom.github.io/Healthwiz-Kingdom

1. **Scan** or open the link: any modern browser, nothing to install.
2. Create your traveller in the **Traveller's Registry**, then log one thing.
3. Tap **Install app** or **Add to Home Screen** to play offline.

GROUP 14
Kolej MARA Kulim

RAYYAN RAFIQIN	QAYYUM
NAUFAL	AMIR
MUI'IZUDDIN	MUIZ ZIKRI

Your data stays on your device. A free account is optional, for backup and sync only.

Wellness tracker, not a medical device. HealthWiz does not diagnose anything and gives no medical advice. If you feel unwell, talk to a parent, teacher or healthcare professional. In an emergency call 999 (Malaysia).



TRACK

THE HEALTH HALL · EIGHT TRACKERS

WATER QUEST Daily goal from your weight (35 mL/kg, the team's own rule). The knight walks, drinks and pours as you log; opt-in live weather adjusts your water potion.

NUTRITION Log from the Kolej MARA Kulim canteen menu. Signed in, **Luke** searches the web for any food, with a source for every part and an energy check (4·protein + 4·carb + 9·fat). Estimates are labelled.

SLEEP Hours against National Sleep Foundation ranges, a Sleep Score (our own game formula), Dream Battle, Counting Sheep and a wake-up alarm.

STRESS QUEST DASS-21 check-in (no severity labels), the Storm Within, Calming Grove breathing, five relief activities. Crisis numbers are always on screen.

BODY & ENERGY BMI (a screening number, not a diagnosis), BMR by Mifflin–St Jeor, daily energy = BMR × FAO/WHO/UNU activity level, macro and fibre goals.

TRAINING HALL 22 moves, 19 muscle maps; Calisthenics is bodyweight only. "I feel dizzy" and "Chest pain" stop buttons always at hand.

RUNNING ROAD GPS runs: distance, pace, calories (MET × kg × h, 2024 Compendium), opt-in Runners' Board.

STAIRS & HEART RATE BPM before, after and 1 minute after a climb for recovery. Camera pulse (video never stored) or a smartwatch / chest strap over Web Bluetooth.

EVERYWHERE Log for yesterday or any earlier date, browse every entry by day, print a weekly report.



PLAY & GUIDE

YOUR LOGS RESTORE THE KINGDOM



THE LIVING KINGDOM Eight regions (Dream Realm, Stair Mountain, Energy Forge, Mind Forest, Heartstone Hall, Water Valley, Nutrition Village, Balance Tower), each in one of five states up to **Flourishing**.

QUESTS & REWARDS Adaptive daily and weekly quests, fair XP with a daily logging allowance, audited badges, gentle streaks, and mini-games for each log.

HEALTH SCORE, 0-100 Five weighted indicators. The **Health Orb** turns it into ranked quest cards: Start Here, Next Step, Keep Going.

MEDIUS NOTICES Insights from your own baselines, a what-if simulator, week-on-week stats and the **Health Time Machine**.

WIZARD'S COUNSEL Talk with Medius or Luna (Gemini, through a secure server function). Signed-in only, rate-limited, with safety rules; nothing is stored. **Ask Luna** finds any feature without AI.

ARCANA The Oracle's crystal ball, Guild War of Joules and Dueling Arena: race a replay of your own earlier climb.

OPT-IN COMMUNITY Hall of Heroes (game progress only) and Runners' Board (nickname and weekly totals only).

FOR EVERYONE 8 themes, performance modes, reduced motion, keyboard and screen-reader friendly charts.



PROVE IT

SCIENCE YOU CAN CHECK

CAMPUS OBSERVATORY A 3D map of Kolej MARA Kulim from OpenStreetMap, in Realistic and Fantasy looks. Each staircase is a heart coloured Mild, Moderate or Vigorous: tap to log a climb or workout. GPS position is never saved or sent.

STAIR SENSE & SIGHT

$h = n \times r$ $W = m \cdot g \cdot h$ $P = W \div t$

Camera and WebXR measure a real staircase's rise and tread.

THE GROUP 14 STAIR EXPERIMENT

Six participants climbed measured stairways. The pulse is modelled as **periodic motion** (f, T, ω) and VO_2 estimated with the project's method. Automated tests **recompute the report's tables from the raw counts**. If the pulse after is lower than before, the app says "not valid" instead of showing a number.

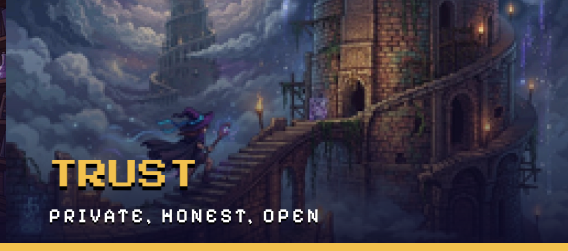
EVIDENCE VAULT Every source on a shelf: claim, how HealthWiz uses it, where you see it. Every "?" links to its entry.

SCIENCE LABORATORY Be Your Own Experiment, the Evidence Trial, and the **Uncertainty Engine** showing how unmeasured assumptions move an estimate.

ASCENT MONUMENT Your climbing builds an endless Tower of Babel, one stone per 0.25 m.

ALCHEMIST'S WORKSHOP Food energy, work against gravity and metabolic use are three different quantities; it shows why.

Sources: Mifflin–St Jeor · FAO/WHO/UNU · 2024 Compendium of Physical Activities · Keytel · Tanaka · Lovibond & Lovibond (DASS-21) · National Sleep Foundation · WHO · IOM. Where a rule is the team's own, the "?" says so.



TRUST

PRIVATE, HONEST, OPEN

YOUR DATA STAYS YOURS

- Saved **on your device**; download or restore a backup any time.
- A free account is **optional**: cloud backup and sync. Row Level Security means only you can read your row.
- **Delete account and cloud data** in two taps.
- Location only **when you tap**, never saved or sent. Camera pulse video never stored.
- AI chat is **never stored**; no AI key inside the app.

UNDER THE HOOD

A static web app: plain JavaScript, no build step. Installable and offline-ready.

PWA · Supabase · Postgres + RLS · Edge Functions · Gemini · MapLibre · Three.js · WebXR · Web Bluetooth · Open-Meteo · Blender

CREDITS AND LICENCES

Fonts: Press Start 2P and Pixelify Sans (SIL OFL 1.1). Map data © OpenStreetMap contributors (ODbL). MapLibre GL JS (BSD-3-Clause). Three.js (MIT). Map avatars: Tom, the anime school boy and girl (CC BY 4.0, suzuart). Campus look follows a drone video by @ikraMRahim, used with the creator's permission. Knight 3D base: MakeHuman CC0 assets. Emblems: MARA and Kolej MARA Kulim. All other characters and scenes are original HealthWiz designs.

SAFE AND TESTED

Saved data is versioned and upgraded step by step, with a backup copy made before any upgrade or repair. **102** browser test files check every feature, with performance budgets and a device matrix. The **Complete Feature Atlas** lists every module.